

# Matplotlib

- Jake VanderPlas. 2016. *Python Data Science Handbook: Essential Tools for Working with Data*. O'Reilly Media, Inc.
- 4. kapitulua - Datuen bistaratzea Matplotlib-ekin
- <https://github.com/jakevdp/PythonDataScienceHandbook>
- **OHARRA:** kode-adibide asko zaharkituta daude

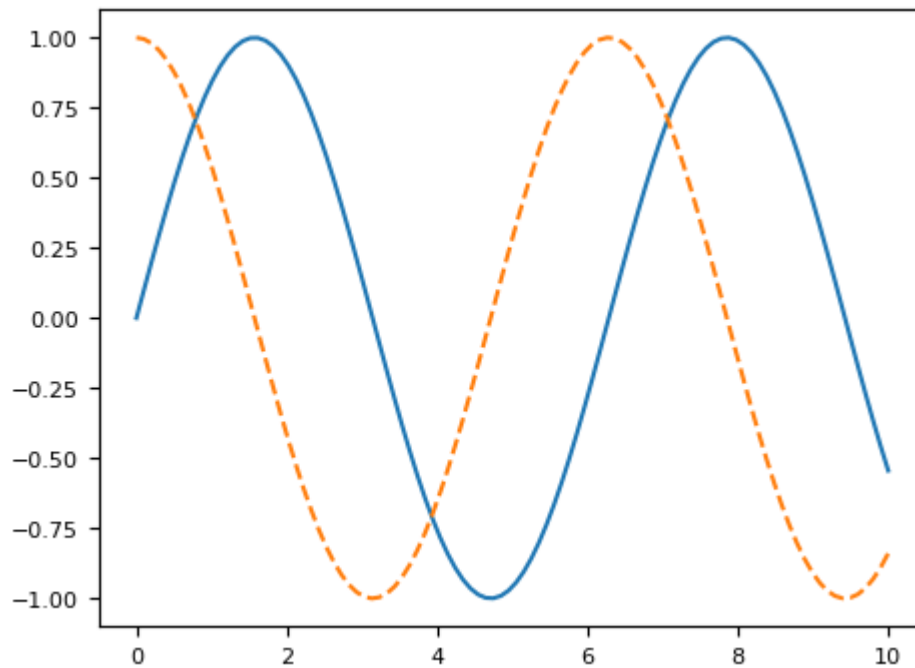
Matplotlib-ek honako hauek eskaintzen ditu:

- Grafiko mota erraz eta pertsonalizagarriak (lerroak, sakabanatze-diagramak, barrak, histogramak, sestra-grafikoak, 3D...)
- NumPy eta Pandas-ekin integrazioa
- Beste liburutegi batzuk erabiliz aberastu daiteke daiteke (Seaborn edo Plotly)
- Irteera-formatu desberdinak
  - Estatikoa ( `%matplotlib inline` ): PNG, JPG, SVG, PDF
  - Interaktiboa ( `%matplotlib notebook` ): Qt, Tkinter and WebAgg backend-ak

```
In [1]: import matplotlib as mpl
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import pandas as pd

# Ezarri irudiaren tamaina Lehenetsia hazbetetan (zabalera, altuera)
mpl.rcParams['figure.figsize'] = (5.33,4)
# Ezarri ardatzen eta marka-etiketen Letra-tamaina Lehenetsia puntutan
mpl.rcParams['axes.labelsize'] = 10 # Adibidea: 14 puntu
mpl.rcParams['xtick.labelsize'] = 8 # Adibidea: 12 puntu x ardatzeko marketarako
mpl.rcParams['ytick.labelsize'] = 8 # Adibidea: 12 puntu y ardatzeko marketarako
```

```
In [2]: x = np.linspace(0, 10, 100)
fig = plt.figure()
plt.plot(x, np.sin(x), '-')
plt.plot(x, np.cos(x), '--');
```



## Interfaze bikoitza

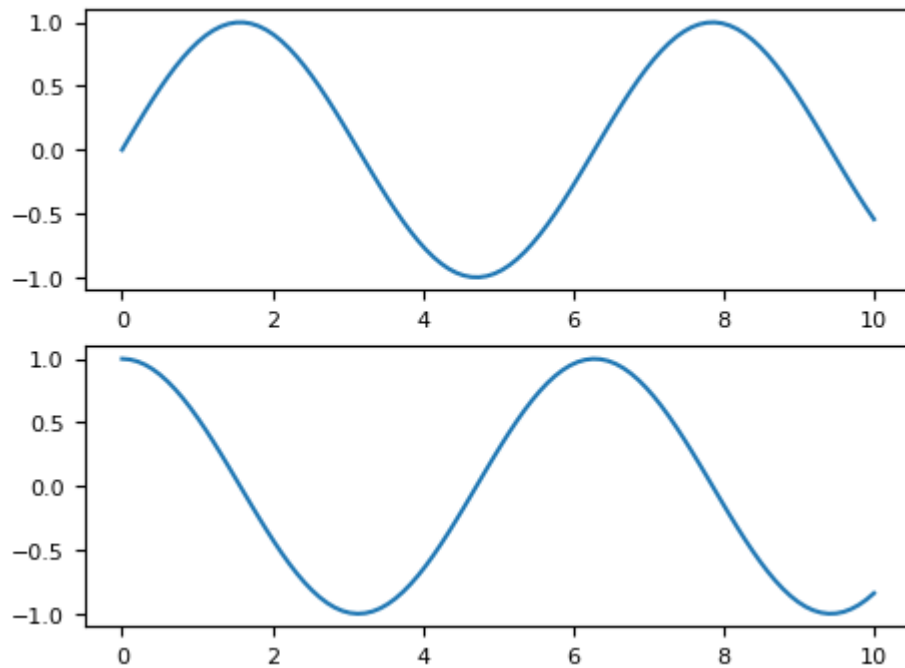
- MATLAB estiloko interfazea
  - Matplotlib: MATLAB-en Python alternatiba
  - `plt` ia guztirako erabili
  - Egokia grafiko sinpleetarako
- Objektuetara bideratutako interfazea
  - Objektuak grafikoak sortu eta haien metodoak deitu
  - Malguagoa

MATLAB estiloko interfazea:

```
In [4]: plt.figure() # sortu grafikoaren irudia

# sortu bi paneletatik lehena eta ezarri uneko ardatza
plt.subplot(2, 1, 1) # (errenkadak, zutabeak, panel-zenbakia)
plt.plot(x, np.sin(x))

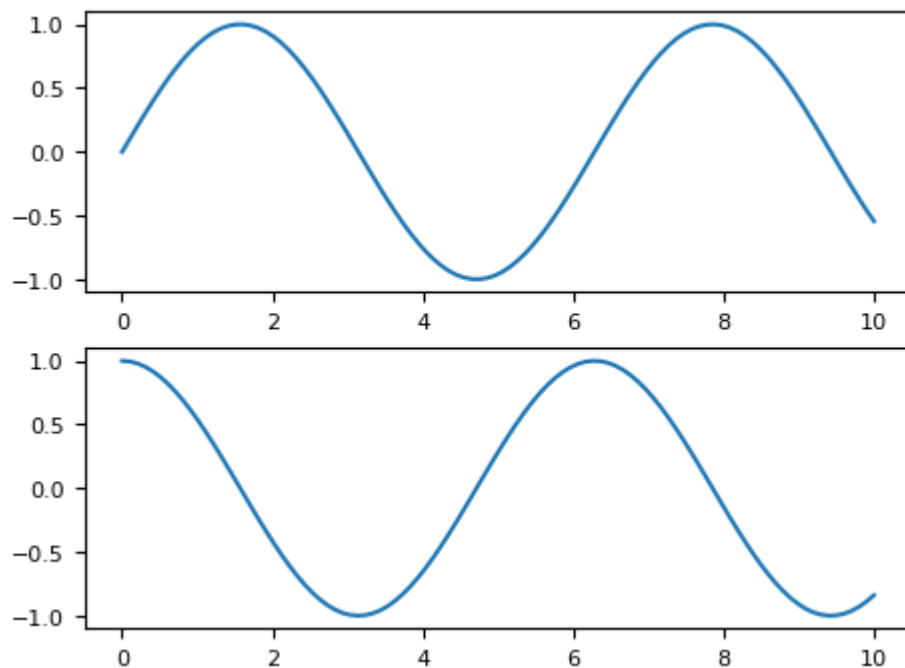
# sortu bigarren panela eta ezarri uneko ardatza
plt.subplot(2, 1, 2)
plt.plot(x, np.cos(x));
```



Objektuetara bideratutako interfazea:

```
In [5]: # Lehenik, grafiko-sareta bat sortu
# ax bi Axes objektutako arraya izango da
fig, ax = plt.subplots(2,1)

# plot() metodoa exekutatu dagokion objektuan
ax[0].plot(x, np.sin(x))
ax[1].plot(x, np.cos(x));
```



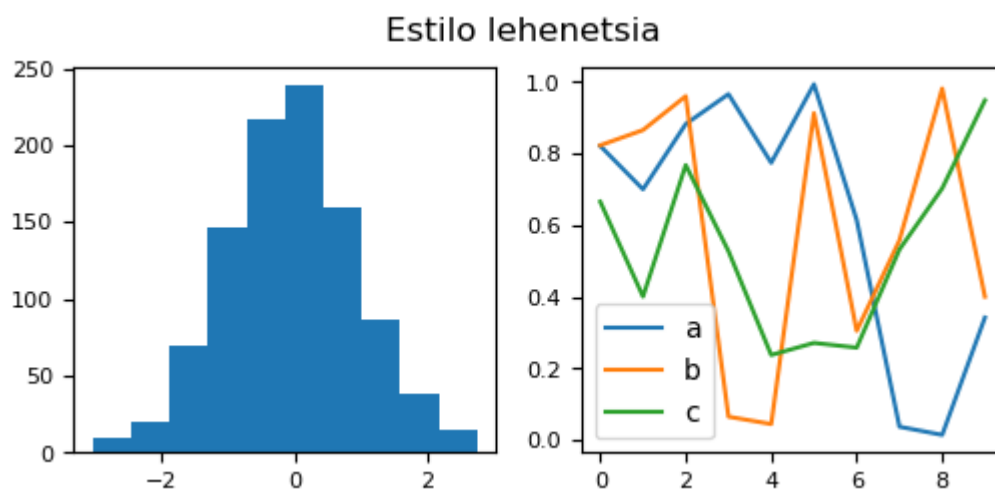
## Estiloak

- Aurrez definitutako atributu bisualen bilduma
- `plt.style.library` → eskuragarri dauden estiloak eta haien parametroak

- `plt.style.available == sorted(plt.style.library.keys())`
- `plt.style.use('stylename')` → estilo globala ezarri
  - Testuinguru-kudeatzailea ( `with context manager`) erabili estiloa lokalki aldatzeko:
  - `with plt.style.context('stylename'):`  
`make_a_plot()`

```
In [6]: def hist_and_lines(title="Estilo lehenetsia"):
        np.random.seed(0)
        fig, ax = plt.subplots(1, 2, figsize=(6, 2.5))
        ax[0].hist(np.random.randn(1000))
        for i in range(3):
            ax[1].plot(np.random.rand(10))
        ax[1].legend(['a', 'b', 'c'], loc='lower left')
        fig.suptitle(title)
```

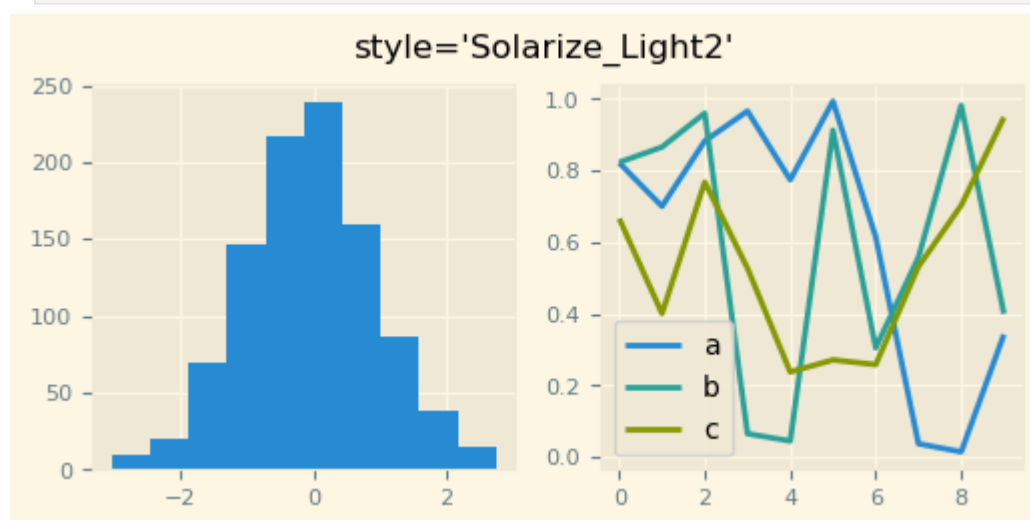
```
In [7]: hist_and_lines()
```



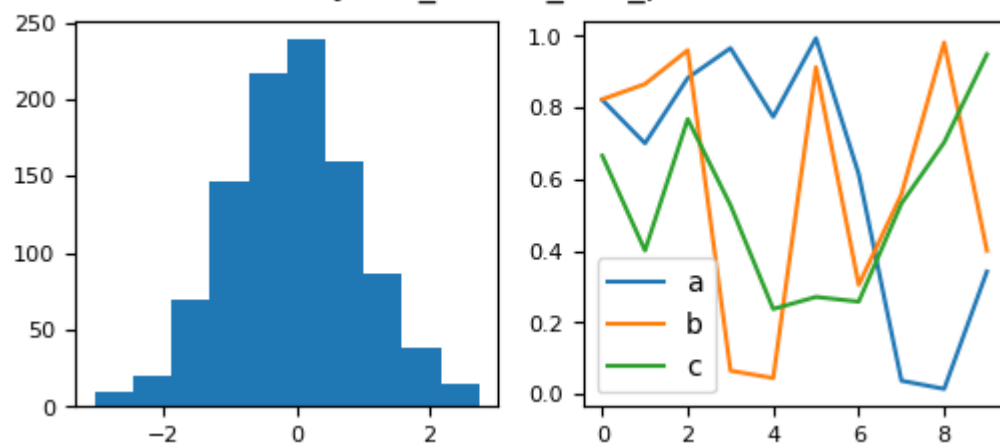
```
In [8]: plt.style.available
```

```
Out[8]: ['Solarize_Light2',
         '_classic_test_patch',
         '_mpl-gallery',
         '_mpl-gallery-nogrid',
         'bmh',
         'classic',
         'dark_background',
         'fast',
         'fivethirtyeight',
         'ggplot',
         'grayscale',
         'petroff10',
         'seaborn-v0_8',
         'seaborn-v0_8-bright',
         'seaborn-v0_8-colorblind',
         'seaborn-v0_8-dark',
         'seaborn-v0_8-dark-palette',
         'seaborn-v0_8-darkgrid',
         'seaborn-v0_8-deep',
         'seaborn-v0_8-muted',
         'seaborn-v0_8-notebook',
         'seaborn-v0_8-paper',
         'seaborn-v0_8-pastel',
         'seaborn-v0_8-poster',
         'seaborn-v0_8-talk',
         'seaborn-v0_8-ticks',
         'seaborn-v0_8-white',
         'seaborn-v0_8-whitegrid',
         'tableau-colorblind10']
```

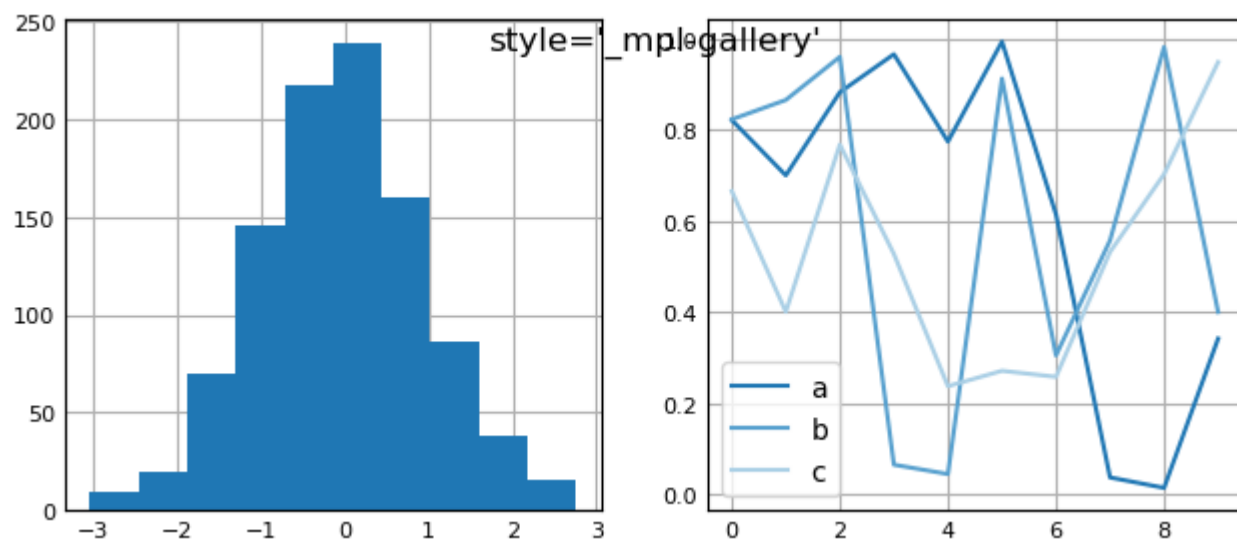
```
In [9]: for style in plt.style.available[:7] :
         with plt.style.context(style):
             hist_and_lines(f'{style=}')
hist_and_lines()
```



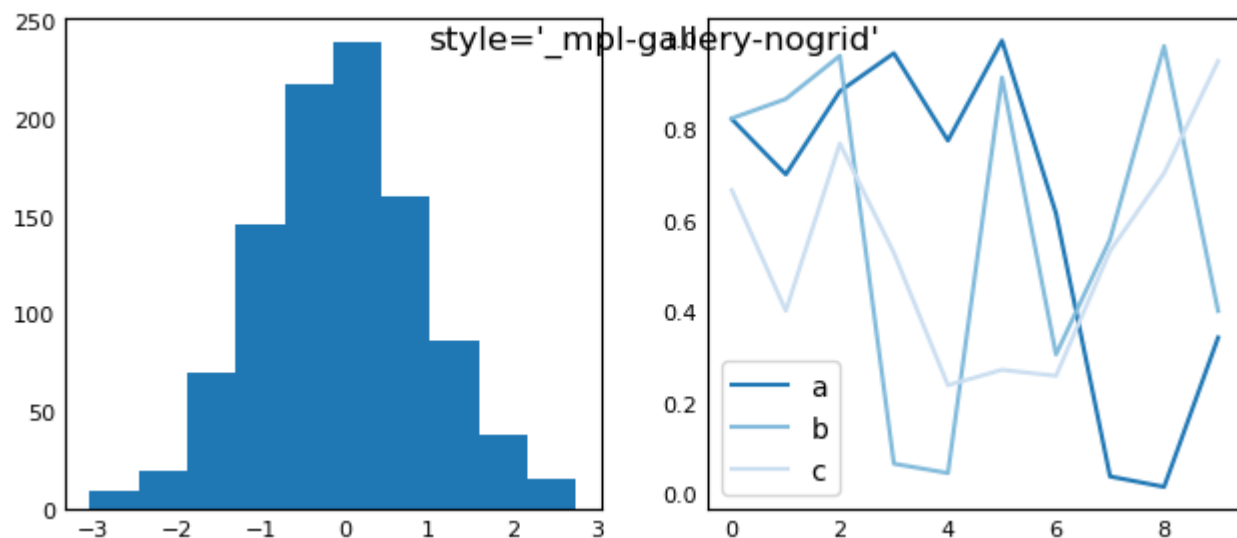
style='\_classic\_test\_patch'



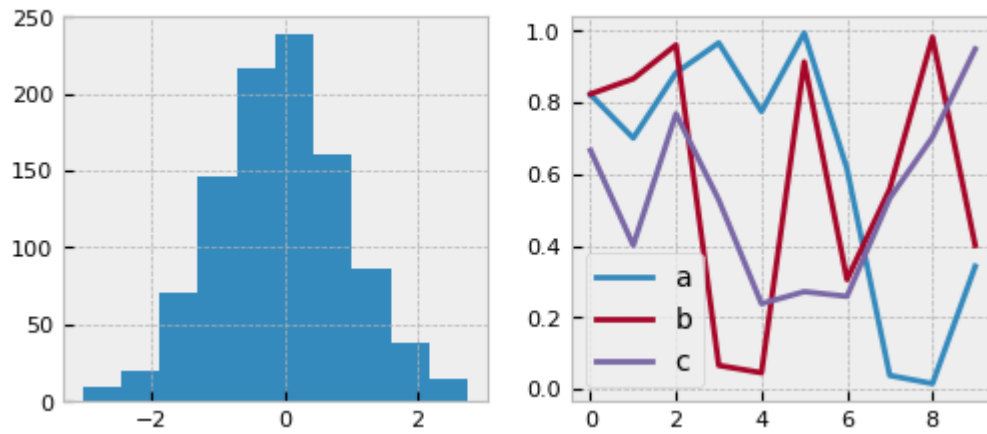
style='\_mpl-gallery'



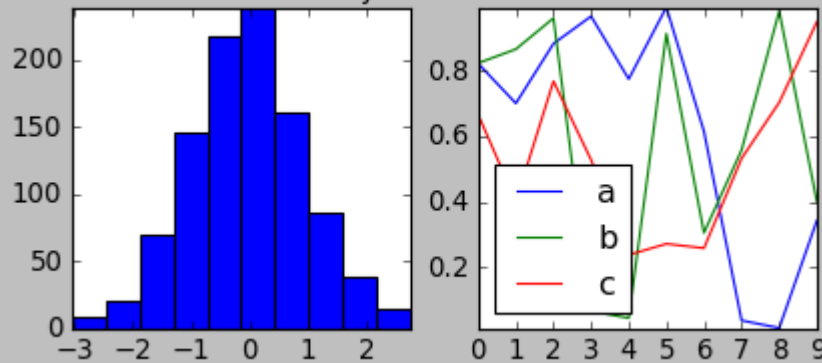
style='\_mpl-gallery-nogrid'



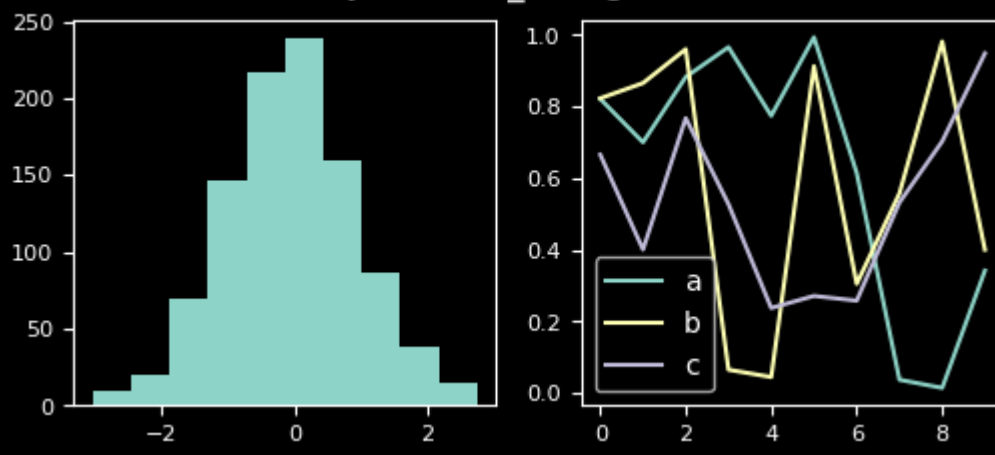
style='bmh'



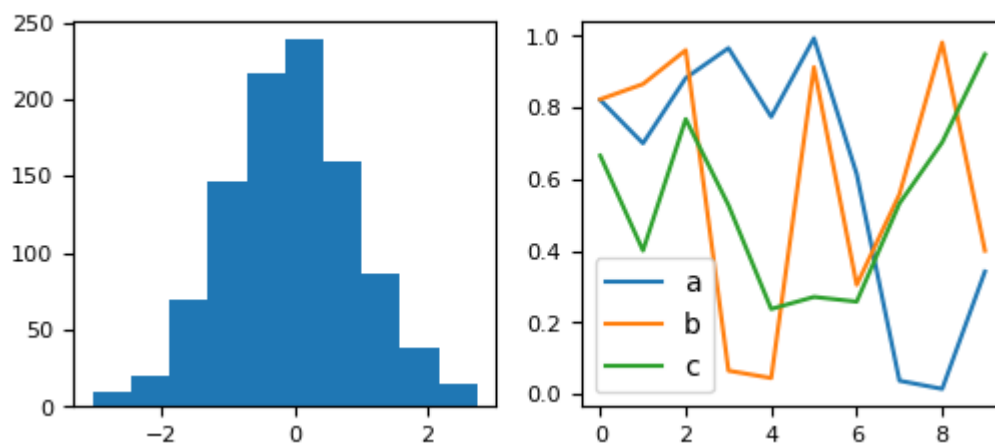
style='classic'



style='dark\_background'



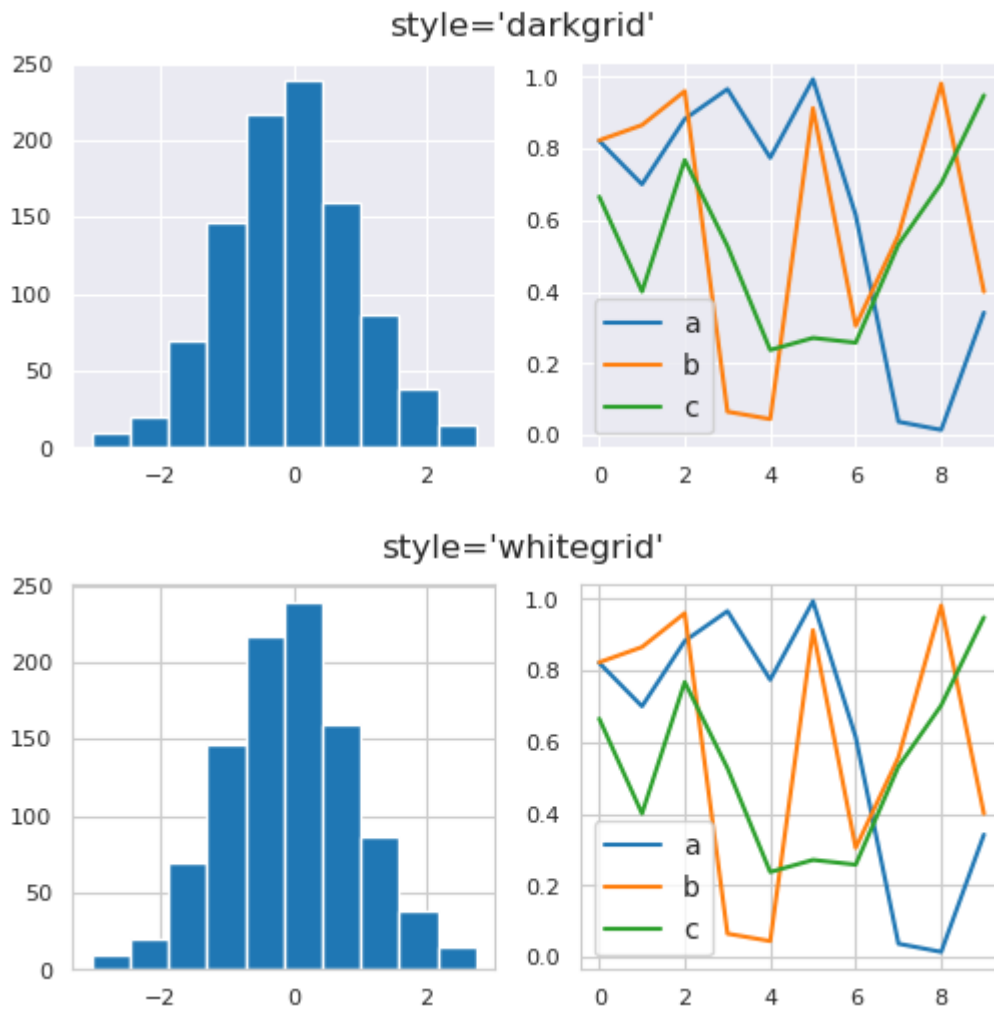
Estilo lehenetsia



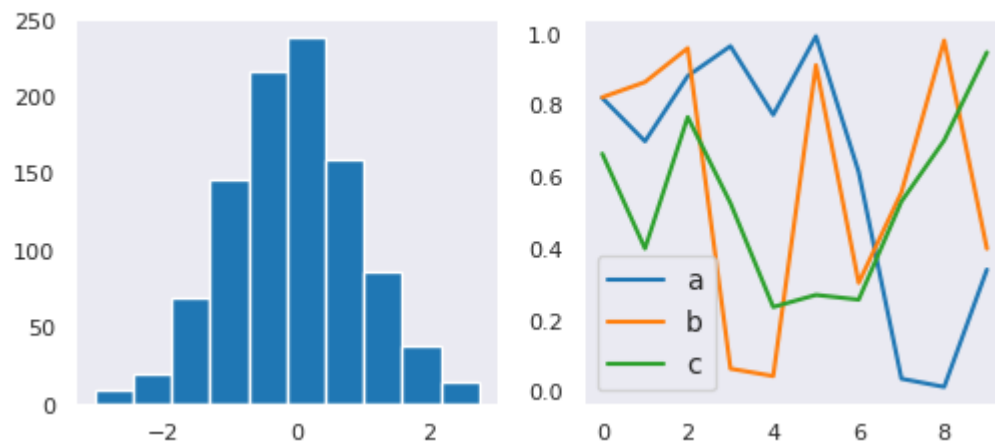
## Seaborn estiloak

- Bost estilo: `darkgrid`, `whitegrid`, `dark`, `white`, and `ticks`
- `sns.set_style(stylename)` → ezarri estilo globala
  - Erabili `testuinguru-kudeatzailea` (`with context manager`) estiloa lokalki aldatzeko:
  - `with sns.axes_style(stylename):`  
`make_a_plot()`

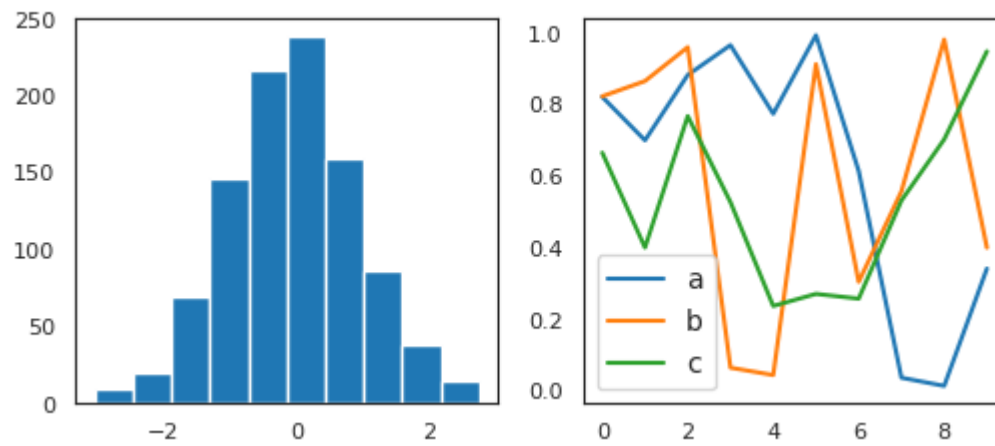
```
In [11]: for style in ['darkgrid', 'whitegrid', 'dark', 'white', 'ticks'] :  
        with sns.axes_style(style):  
            hist_and_lines(f'{style=}')  
hist_and_lines()
```



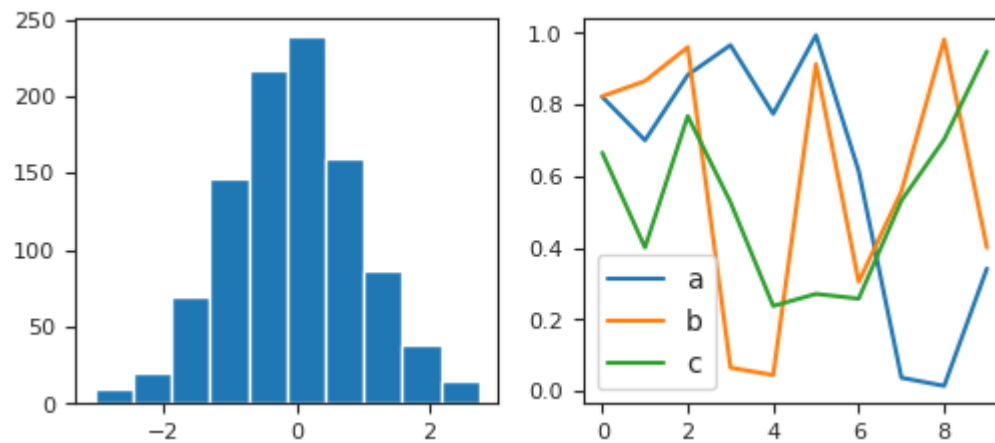
style='dark'



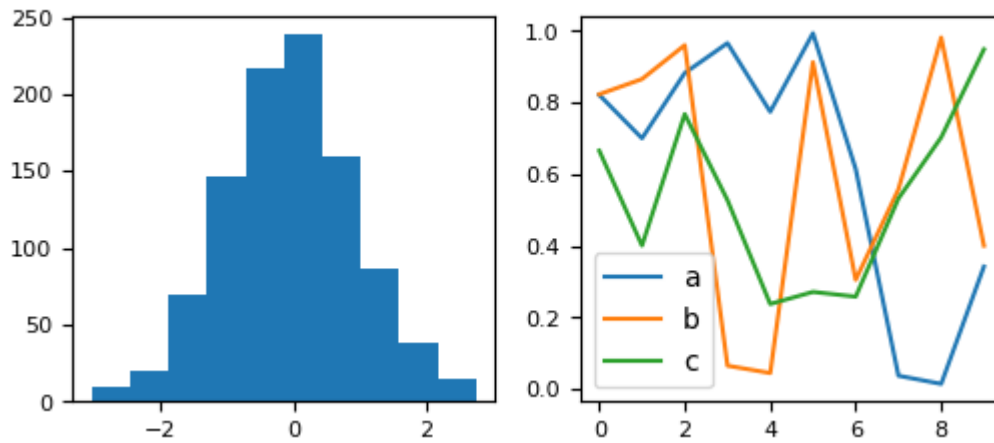
style='white'



style='ticks'



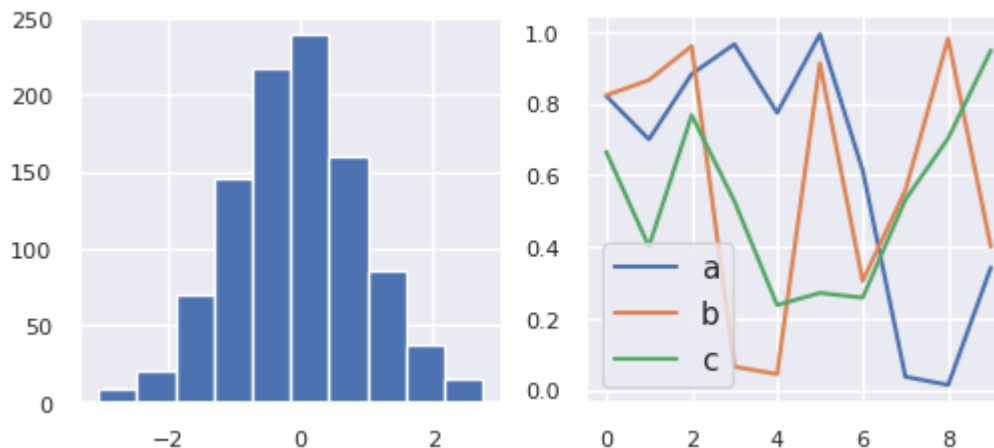
## Estilo lehenetsia



- `sns.set()` == `sns.set_theme()` → `sns.set_style('darkgrid')` eta gehiago

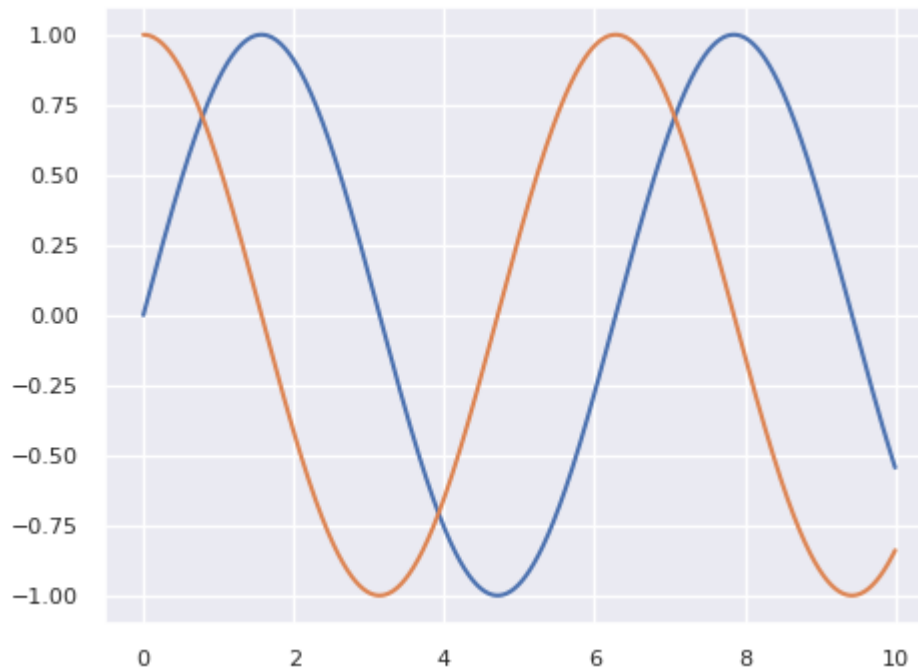
```
In [12]: sns.set()
mpl.rcParams['figure.figsize'] = (5.33,4)
mpl.rcParams['axes.labelsize'] = 10 # Adibidea: 14 puntu
mpl.rcParams['xtick.labelsize'] = 8 # Adibidea: 12 puntu x ardatzeko marketarako
mpl.rcParams['ytick.labelsize'] = 8 # Adibidea: 12 puntu y ardatzeko marketarako
hist_and_lines("Estilo lehenetsi berria")
```

## Estilo lehenetsi berria

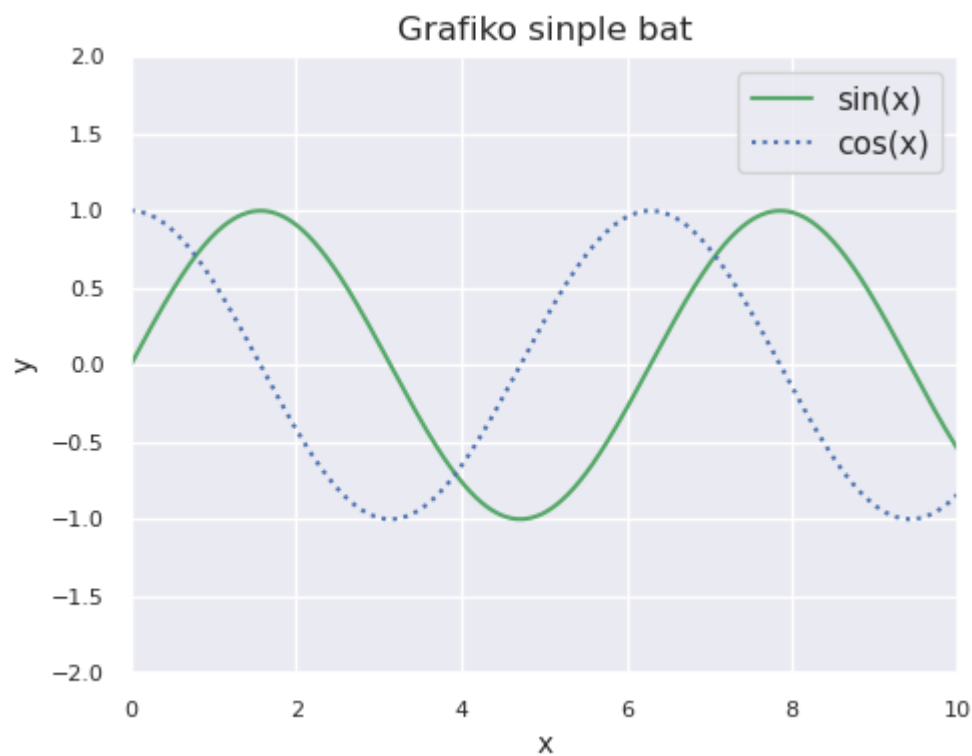


## Lerroen, ardatzen eta etiketen propietateak

```
In [13]: x = np.linspace(0, 10, 1000)
ax = plt.axes()
ax.plot(x, np.sin(x))
ax.plot(x, np.cos(x));
```



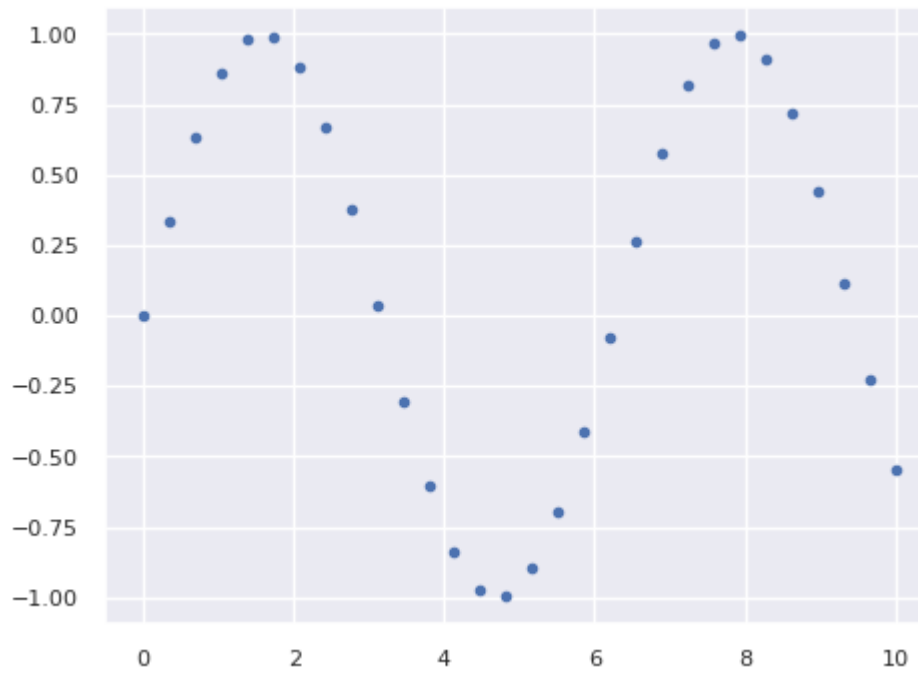
```
In [14]: x = np.linspace(0, 10, 1000)
ax = plt.axes()
ax.plot(x, np.sin(x), '-g', label='sin(x)')
ax.plot(x, np.cos(x), ':-b', label='cos(x)')
ax.set(xlim=(0, 10), ylim=(-2, 2),
       xlabel='x', ylabel='y',
       title='Grafiko sinple bat')
plt.legend();
```



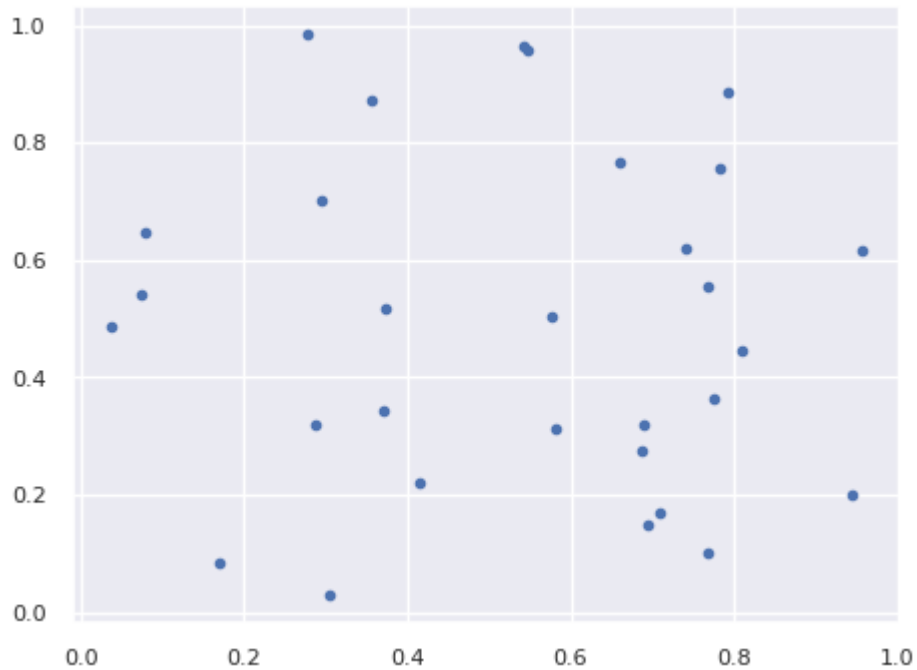
## Sakabanatze-diagramak

```
In [15]: x = np.linspace(0, 10, 30)
y = np.sin(x)
```

```
#plt.plot(x, y);      # lehenetsia: '-' lerroak erabili
plt.plot(x, y, '.'); # erabili puntua markatzaile gisa
```



```
In [16]: x,y = np.random.rand(2,30)
#plt.plot(x, y);      # lehenetsia: '-' lerroak erabili
plt.plot(x, y, '.'); # erabili puntua markatzaile gisa
```



- `scatter(x, y, s=, c=, cmap=)` → puntu bakoitzerako tamaina eta kolorea dituen sakabanatze-diagrama

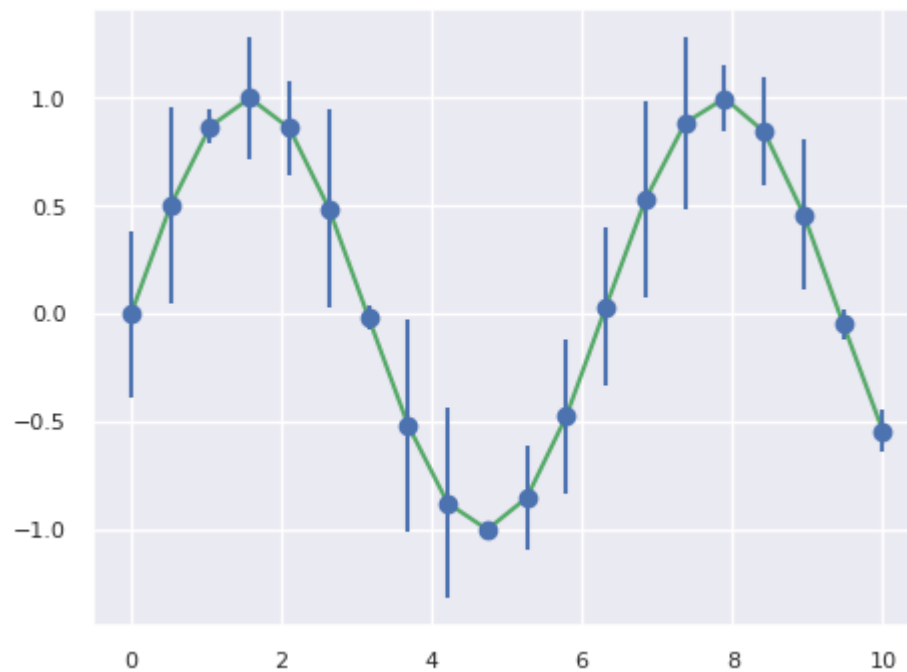
```
In [17]: x,y,sizes,colors = np.random.rand(4,30)
plt.scatter(x, y, s=sizes*1000, alpha=0.3, c=colors, cmap='viridis')
plt.colorbar(); # erakutsi kolore-eskala
```



## Errore-barrak

- `plt.errorbar(x, y, yerr=)` → errore-barren grafiko bat

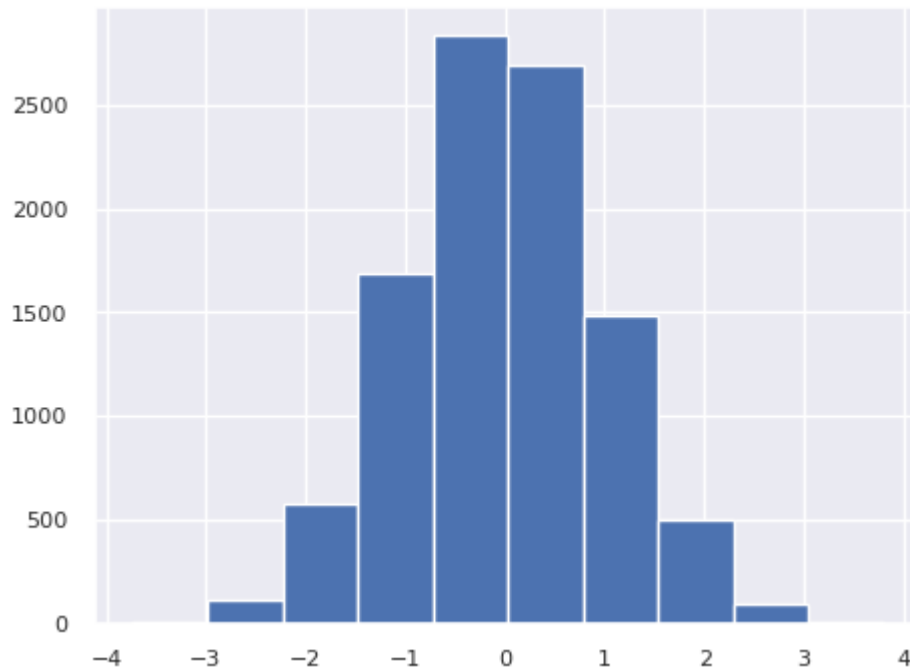
```
In [18]: x = np.linspace(0, 10, 20)
y = np.sin(x)
dy = 0.5 * np.random.rand(x.size)
plt.plot(x, np.sin(x), 'g')
plt.errorbar(x, np.sin(x), yerr=dy, fmt='o');
```



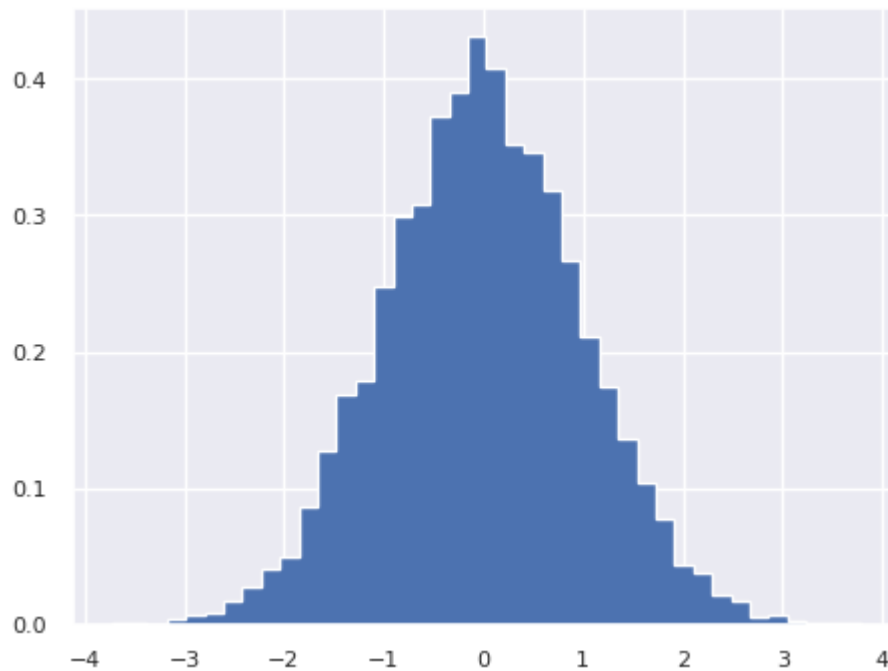
## Histogramak

- `plt.hist(x, bins=, density=, histtype=)` → histograma bat

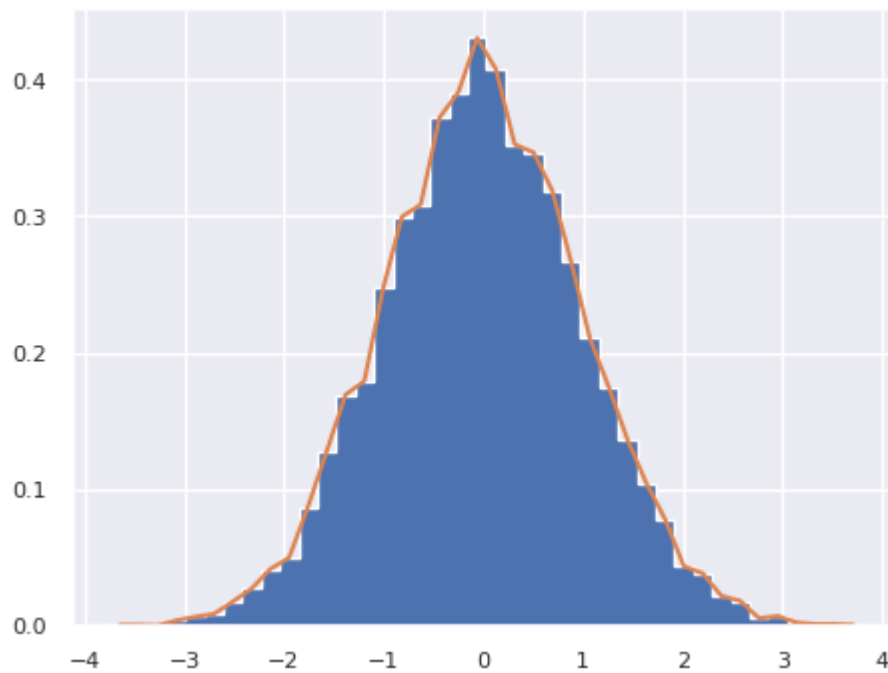
```
In [19]: x = np.random.randn(10000)
plt.hist(x);
```



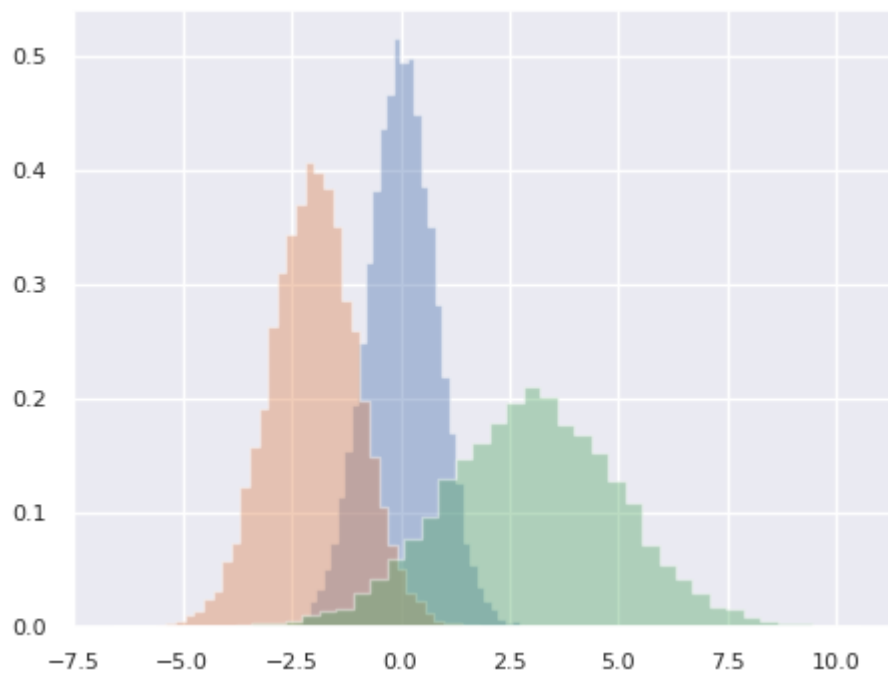
```
In [20]: plt.hist(x, bins=40, density=True, histtype='stepfilled');
```



```
In [21]: density, bins, patches = plt.hist(x, bins=40, density=True, histtype='stepfilled')
# Kalkulatu tarteen erdigunea
bin_centers = (bins[:-1] + bins[1:]) / 2
# Marraztu probabilitate-dentsitatea lerro gisa
plt.plot(bin_centers, density);
```



```
In [22]: x1 = np.random.normal(0, 0.8, 10000)
x2 = np.random.normal(-2, 1, 10000)
x3 = np.random.normal(3, 2, 10000)
kwargs = dict(bins=40, density=True, histtype='stepfilled', alpha=0.4)
plt.hist(x1, **kwargs)
plt.hist(x2, **kwargs)
plt.hist(x3, **kwargs);
```



## Seaborn

- Matplotlib Pandas baino hamarkada bat lehenago sortu zen
  - Ez zen Pandas-ekin erabiltzeko diseinatu (eguneratzen ari da)
- Seaborn-ek Matplotlib-en gaineko API bat eskaintzen du eta Pandas-ekin integratzen da
  - Tresna estatistikoak: erregresioak, erroreak eta banaketak

■ Aldagai anitzeko bistaratze sinplea

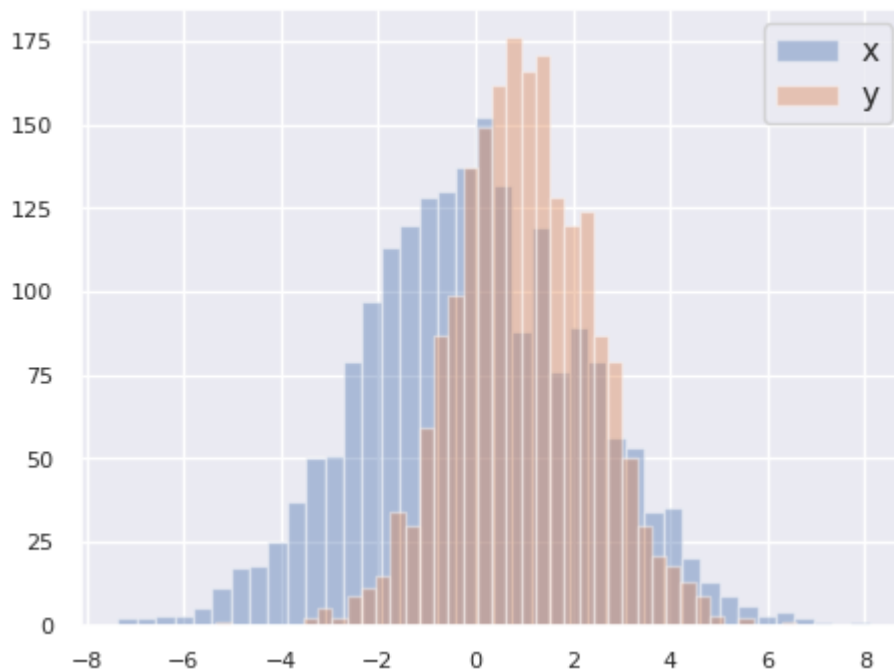
```
In [23]: a = np.random.multivariate_normal(mean=[0, 1], cov=[[5, 2], [2, 2]], size=2000)
df = pd.DataFrame(a, columns=['x', 'y'])
df.head()
```

```
Out[23]:
```

|   | x         | y        |
|---|-----------|----------|
| 0 | 0.422732  | 0.152363 |
| 1 | -0.733416 | 0.678723 |
| 2 | -1.825696 | 0.841521 |
| 3 | 2.113652  | 1.340771 |
| 4 | 1.254496  | 1.035458 |

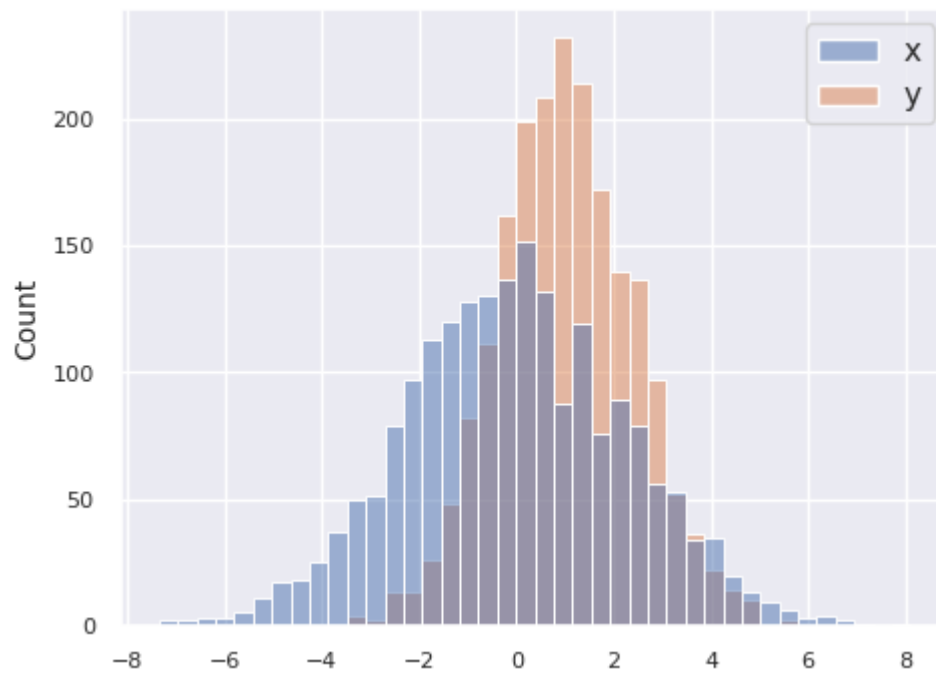
Bi histograma marraztea (tarte-zabalerak desberdinak dira)

```
In [24]: plt.hist(df['x'], bins=40, alpha=0.4, label='x')
plt.hist(df['y'], bins=40, alpha=0.4, label='y')
plt.legend();
```



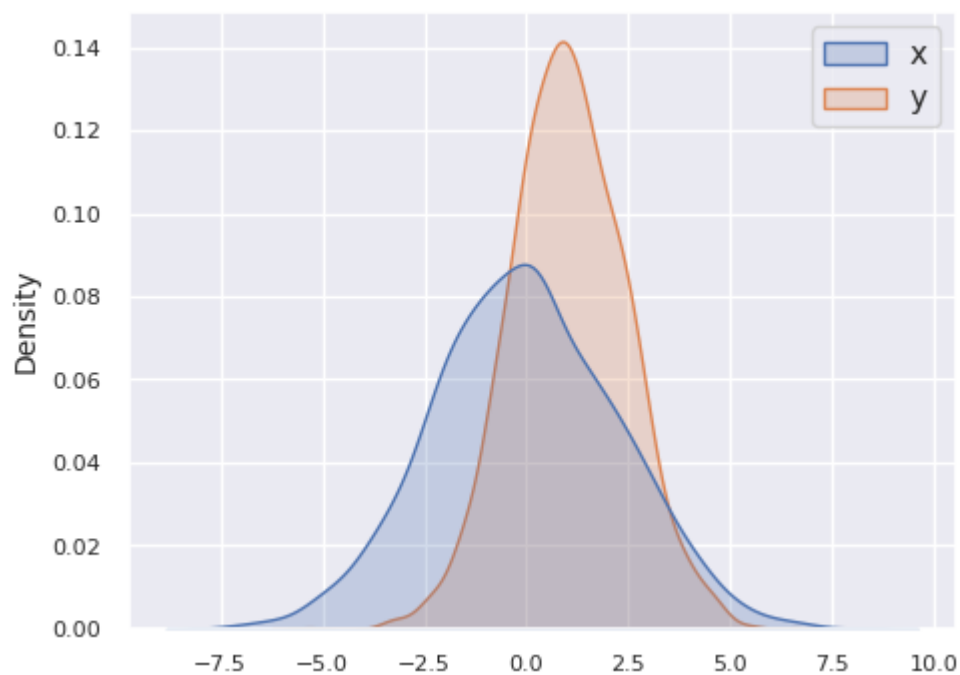
Seaborn-ek DataFrame-ko aldagaien histogramak marraz ditzake (tarte-zabalera berdinekin):

```
In [25]: sns.histplot(df, bins=40);
```



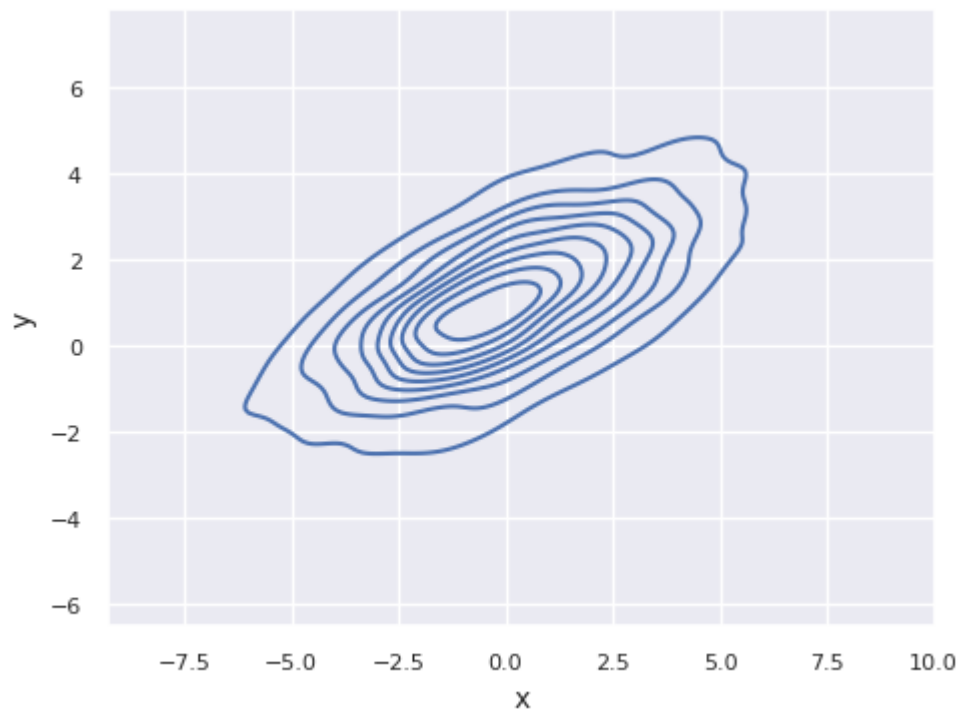
DataFrame-ko aldagaien kernel-dentsitatearen estimazioa:

```
In [26]: sns.kdeplot(df, fill=True);
```



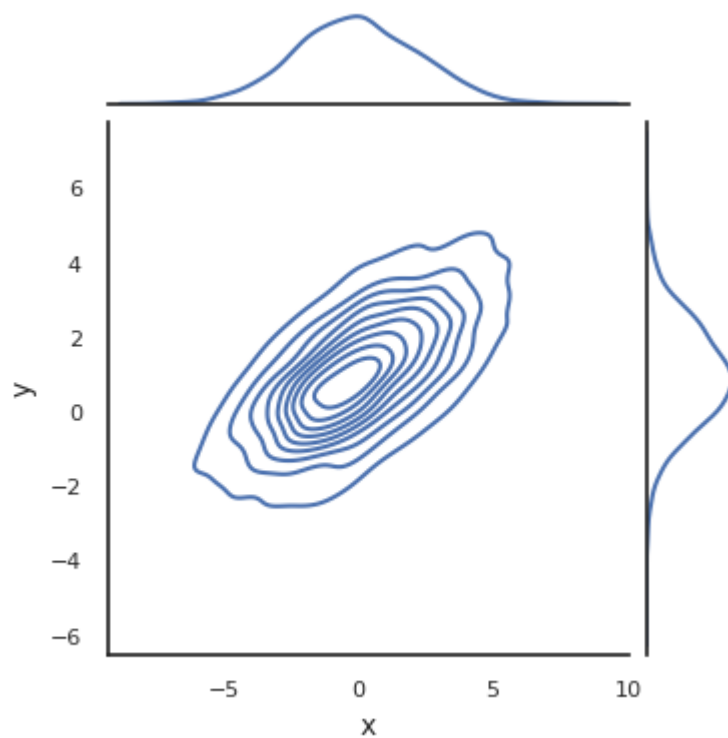
Kernel-dentsitatearen estimazioaren bi dimentsioko bistaratzea:

```
In [27]: sns.kdeplot(df, x='x', y='y');
```



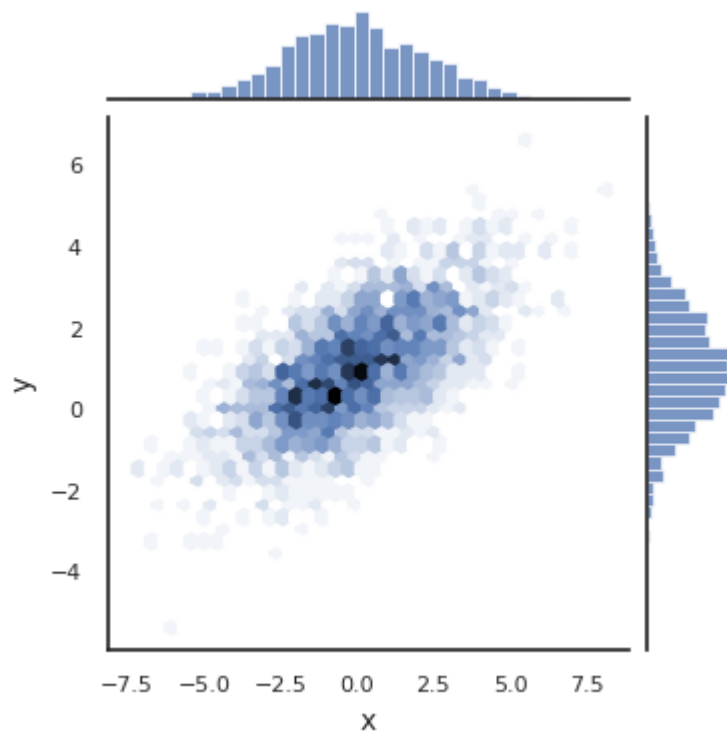
Banaketa bateratua eta marjinalak batera:

```
In [28]: with sns.axes_style('white'):
sns.jointplot(df, x='x', y='y', kind='kde', height=4);
```



Gauza bera, baina hexagonoetan oinarritutako histograma erabiliz:

```
In [29]: with sns.axes_style('white'):
sns.jointplot(df, x='x', y='y', kind='hex', height=4);
```



## Aldagai anitzeko bistaratzea

**Iris** datu multzoa: hiru iris-espezieren petaloen eta sepaloen neurketak:

```
In [30]: iris = sns.load_dataset("iris")
print(iris.shape)
iris.head()
```

(150, 5)

```
Out[30]:
```

|   | sepal_length | sepal_width | petal_length | petal_width | species |
|---|--------------|-------------|--------------|-------------|---------|
| 0 | 5.1          | 3.5         | 1.4          | 0.2         | setosa  |
| 1 | 4.9          | 3.0         | 1.4          | 0.2         | setosa  |
| 2 | 4.7          | 3.2         | 1.3          | 0.2         | setosa  |
| 3 | 4.6          | 3.1         | 1.5          | 0.2         | setosa  |
| 4 | 5.0          | 3.6         | 1.4          | 0.2         | setosa  |

Datuen korrelazio-matrizea:

```
In [31]: iris.drop(['species'], axis=1).corr()
```

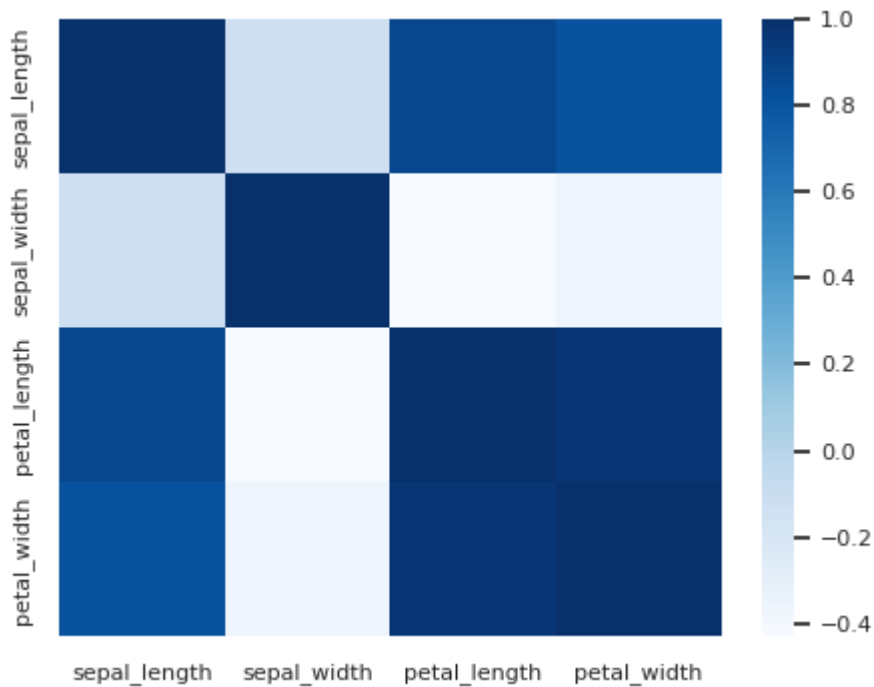
```
Out[31]:
```

|              | sepal_length | sepal_width | petal_length | petal_width |
|--------------|--------------|-------------|--------------|-------------|
| sepal_length | 1.000000     | -0.117570   | 0.871754     | 0.817941    |
| sepal_width  | -0.117570    | 1.000000    | -0.428440    | -0.366126   |
| petal_length | 0.871754     | -0.428440   | 1.000000     | 0.962865    |
| petal_width  | 0.817941     | -0.366126   | 0.962865     | 1.000000    |

Korrelazio-matrizearen bero-mapa:

```
In [32]: c = iris.drop(['species'], axis=1).corr()  
sns.heatmap(c, cmap="Blues")
```

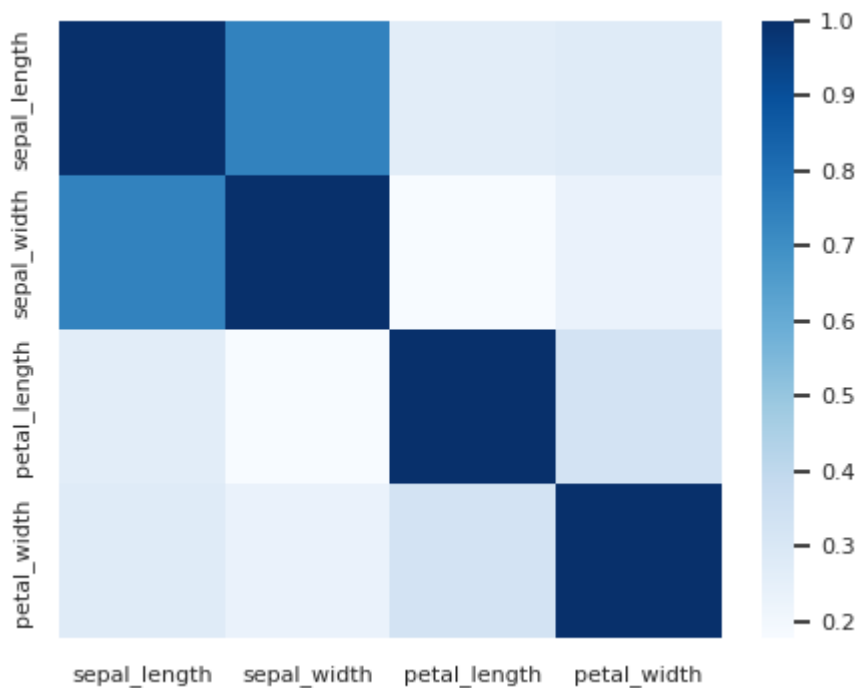
Out[32]: <Axes: >



setosa espeziearen korrelazio-matrizearen bero-mapa:

```
In [33]: c = iris[iris['species']=='setosa'].drop('species', axis=1).corr()  
sns.heatmap(c, cmap="Blues")
```

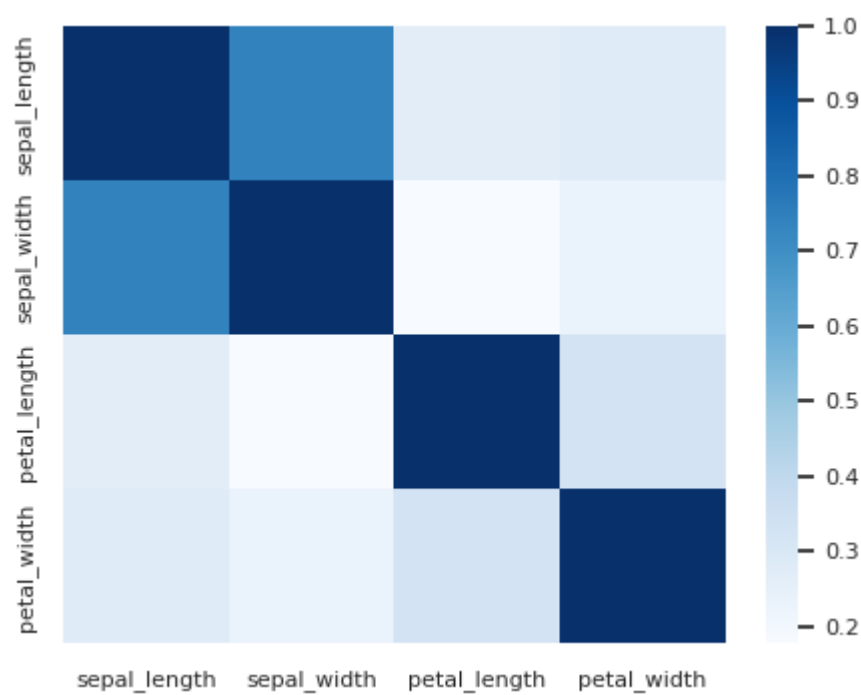
Out[33]: <Axes: >



Gauza bera, baina species aldagaia indize gisa erabiliz:

```
In [34]: c = iris.set_index('species').loc['setosa'].corr()  
sns.heatmap(c, cmap="Blues")
```

Out[34]: <Axes: >



Espezie bakoitzaren korrelazio-matrizea:

```
In [35]: iris.groupby('species').corr()
```

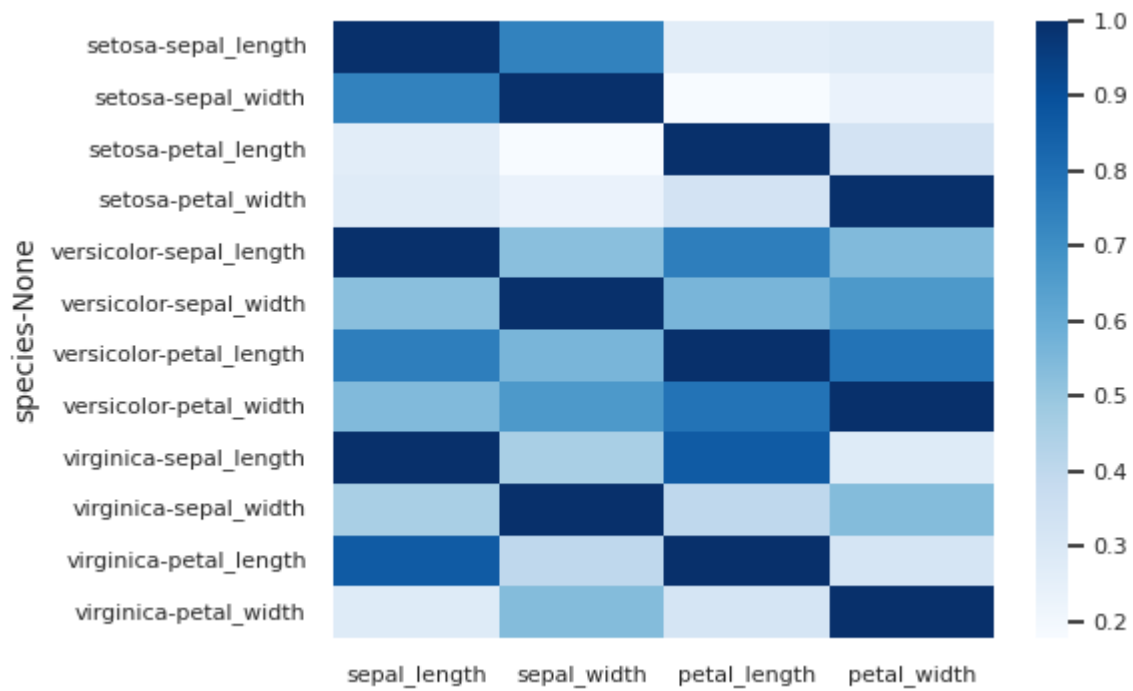
Out[35]:

|            |              | sepal_length | sepal_width | petal_length | petal_width |
|------------|--------------|--------------|-------------|--------------|-------------|
| species    |              |              |             |              |             |
| setosa     | sepal_length | 1.000000     | 0.742547    | 0.267176     | 0.278098    |
|            | sepal_width  | 0.742547     | 1.000000    | 0.177700     | 0.232752    |
|            | petal_length | 0.267176     | 0.177700    | 1.000000     | 0.331630    |
|            | petal_width  | 0.278098     | 0.232752    | 0.331630     | 1.000000    |
| versicolor | sepal_length | 1.000000     | 0.525911    | 0.754049     | 0.546461    |
|            | sepal_width  | 0.525911     | 1.000000    | 0.560522     | 0.663999    |
|            | petal_length | 0.754049     | 0.560522    | 1.000000     | 0.786668    |
|            | petal_width  | 0.546461     | 0.663999    | 0.786668     | 1.000000    |
| virginica  | sepal_length | 1.000000     | 0.457228    | 0.864225     | 0.281108    |
|            | sepal_width  | 0.457228     | 1.000000    | 0.401045     | 0.537728    |
|            | petal_length | 0.864225     | 0.401045    | 1.000000     | 0.322108    |
|            | petal_width  | 0.281108     | 0.537728    | 0.322108     | 1.000000    |

Espezie bakoitzaren korrelazio-matrizearen bero-mapa:

```
In [36]: sns.heatmap(iris.groupby('species').corr(),cmap="Blues")
```

```
Out[36]: <Axes: ylabel='species-None'>
```



Aldagai bikoteen sakabanatze-diagramak eta KDE dituen grafikoa:

```
In [37]: sns.pairplot(iris, hue='species', height=1.5);
```

